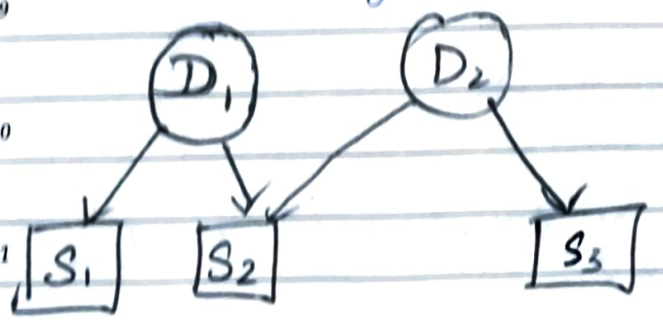
Scale the image.

Bayesian Network:



$P(D_1) = 0.4, \quad P(D_2) = 0.7$

$P(S_1|D_1) = 0.3 \quad P(S_1|D_1') = 0.6$

$P(S_1) = P(S_1|D_1)\,P(D_1) +$

$P(S_1|D_1') * P(S_1|D_1')$

$= (0.3)(0.4) + (0.6)(0.6)$

$= 0.12 + 0.36$

$P(S_1) = 0.48$

Notes

$P(A) \quad P(B)$

$\searrow \quad \swarrow$

$P(C)$

$\downarrow$

$P(D)$

$P(A) = 0.2 \qquad P(A=1) = 0.8$

$P(B=0) = 0.6 \qquad P(B=1) = 0.4$

$P(C|A,B) \Rightarrow$

A	B	C=0	C=1
0	0	0.2	0.8
0	1	0.6	0.4
1	0	0.3	0.7
1	1	0.5	0.5

C	P(D|C) D=0	D=1
0	0.1	0.9
1	0.9	0.1

$P(A=0, B=1, C=0, D=1) \Rightarrow$

$\Rightarrow P(D=1|C=0) \cdot P(C=0|P(A=0; B=1)) \cdot P(A=0) \cdot (P(B=1))$

$\Rightarrow (0.9) \cdot (0.6) \cdot (0.2)(0.4)$

$= 0.0288$

(2) $P(C=0) = 0.5$

$P(S|C)$

C	S=1	S=0
0	0.5	0.5
1	0.1	0.9

Cloudy

Sprinkles

Rain

Wet grass

C	R=1	R=0
1	0.8	0.2
0	0.2	0.8

(ii) $P(i=1, d=1, g=1, s=1, l=1)$

$\Rightarrow P(l=1 \mid g=1) * P(s=1 \mid d=1) * P(g=1 \mid i=1, d=1) * P(i=1) * P(d=1)$

$\Rightarrow (0.8) * (0.8) * (0.5) * (0.4) * (0.4)$

$\Rightarrow 0.0512$

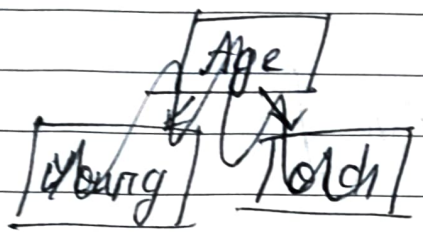
Decision Tree Model:

Age(x_1)	Tumor shape(x_2)	Color x_3	Cancer
1) old	Round(R)	Light	1
2) Young	NR	Light	1
3) Young	R	Dark	0
4) old	NR	Light	0
5) old	R	Light	1
6) old	R	Dark	1
7) Young	NR	Dark	0
8) old	R	Dark	1
9) Young	R	Dark	0
10) Young	R	Dark	0

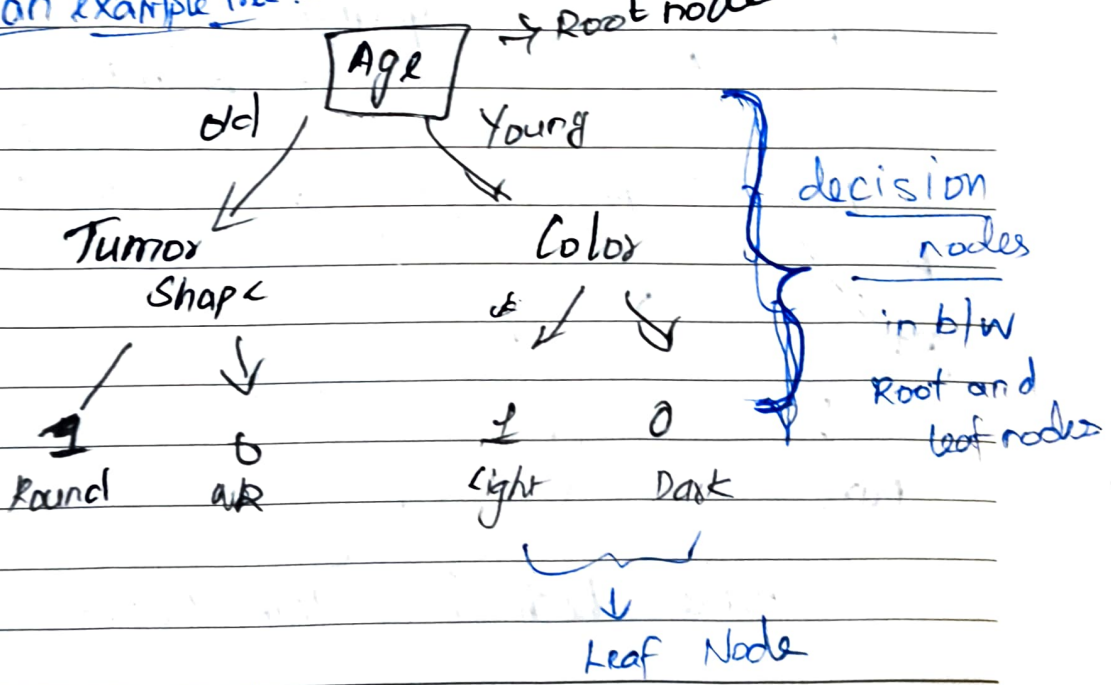
⇒ Binary Classification problem
3 features, 10 examples

are taking discrete values

x_1, x_2, x_3 → more than two possible values.



Just an example tree :-



Notes

⇒ Multiple decision trees possible for a dataset
⇒ Decision tree with best accuracy value is selected.

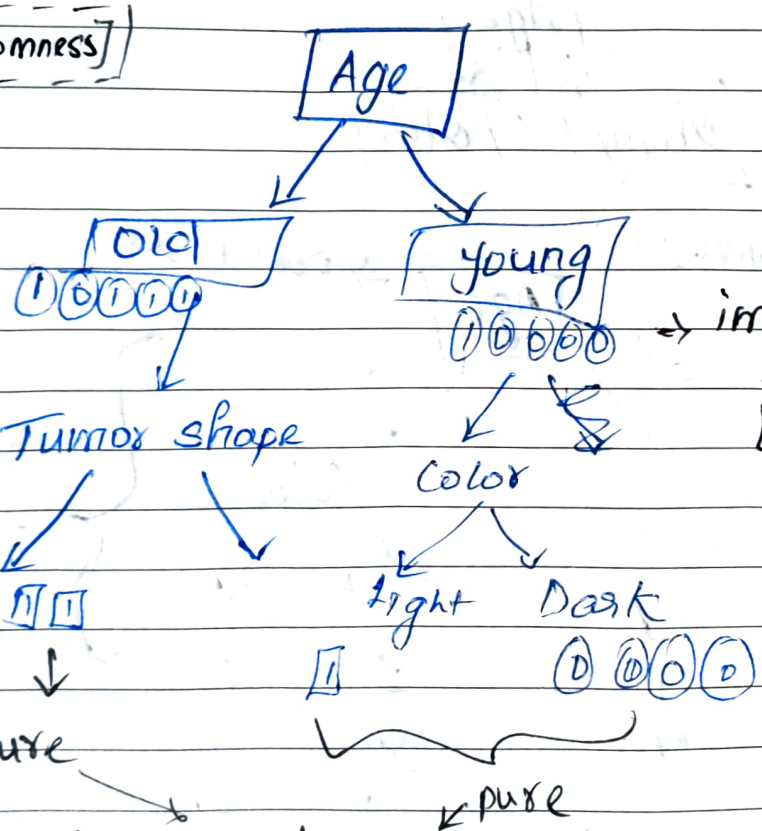
04 March

Tuesday

~~First~~ steps of Decision Tree:-

* Decide feature to use at the ROOT node
* Focus on the left point and decide what node to put below it.

∴ Entropy quantifies impurity [Randomness]



→ impure [So the not leaf node [go further]

∴ Hence both are leaf nodes

Notes